

A STOCHASTIC DERIVATION OF THE BIVARIATE POISSON DISTRIBUTION

M.A. Hamdan

American University of Beirut

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SUMMARY

A bivariate generalization of the univariate homogeneous Poisson process is given. Starting from basic assumptions, we derive a differential equation for the bivariate probability generating function. Hence, the corresponding bivariate probability density function is obtained. This provides an instructive stochastic derivation of the bivariate Poisson distribution.

1. Introduction

The bivariate Poisson distribution can be derived as the limiting form of fourfold sampling with replacement or the limiting form of the bivariate binomial distribution [Campbell (1934), Hamdan and Tsokos (1971)]. In this note, a bivariate form of the homogeneous Poisson process [Doob (1953)] is given. This results in an axiomatic stochastic derivation of the bivariate Poisson distribution. The bivariate Poisson process establishes the basis for the use of the bivariate Poisson distribution as a model for bivariate accident data [Maritz (1950)].

2. Assumptions for the bivariate Poisson process

Consider a process associated with two events A_1 and A_2 such that in any time interval $(t, t + \Delta t)$ either event may or may not occur with the following probabilities:

- (i) $\Pr(A_1 A_2) = \lambda_{12} \Delta t + o(\Delta t),$
- (ii) $\Pr(A_1 \bar{A}_2) = (\lambda_1 - \lambda_{12}) \Delta t + o(\Delta t),$
- (iii) $\Pr(\bar{A}_1 A_2) = (\lambda_2 - \lambda_{12}) \Delta t + o(\Delta t),$
- (iv) $\Pr(\bar{A}_1 \bar{A}_2) = 1 - (\lambda_1 + \lambda_2 - \lambda_{12}) \Delta t + o(\Delta t),$
- (v) $\Pr(\text{more than one occurrence of either } A_1 \text{ or } A_2) = o(\Delta t).$

(vi) Assume further that occurrences in successive intervals (of length Δt) are independent.

λ_1, λ_2 and λ_{12} are constants.

Let $X_1(t), X_2(t)$ be the random variables corresponding to the numbers of occurrences of A_1, A_2 in $(0, t)$. Write $P_{x_1, x_2}(t), x_1, x_2 \geq 0$, for $\Pr[X_1(t)=x_1, X_2(t)=x_2]$, with initial conditions $P_{0,0}(0) = 1, P_{x_1, x_2}(0) = 0$ for $(x_1, x_2) \neq (0, 0)$.

3. Derivation of the bivariate probability density function

By the above assumptions, we have

$$(3.1) \quad P_{0,0}(t+\Delta t) = [1 - (\lambda_1 + \lambda_2 - \lambda_{12}) \Delta t] P_{0,0}(t) + o(\Delta t),$$

$$(3.2) \quad P_{x_1,0}(t+\Delta t) = [1 - (\lambda_1 + \lambda_2 - \lambda_{12}) \Delta t] P_{x_1,0}(t) + (\lambda_1 - \lambda_{12}) \Delta t P_{x_1-1,0}(t) + o(\Delta t), \quad x_1 \geq 1,$$

$$(3.3) \quad P_{0,x_2}(t+\Delta t) = [1 - (\lambda_1 + \lambda_2 - \lambda_{12}) \Delta t] P_{0,x_2}(t) + (\lambda_2 - \lambda_{12}) \Delta t P_{0,x_2-1}(t) + o(\Delta t), \quad x_2 \geq 1,$$

$$(3.4) \quad P_{x_1, x_2}(t+\Delta t) = [1 - (\lambda_1 + \lambda_2 - \lambda_{12}) \Delta t] P_{x_1, x_2}(t) + (\lambda_1 - \lambda_{12}) \Delta t P_{x_1-1, x_2}(t) + (\lambda_2 - \lambda_{12}) \Delta t P_{x_1, x_2-1}(t) + \lambda_{12} \Delta t P_{x_1-1, x_2-1}(t) + o(\Delta t), \quad x_1, x_2 \geq 1.$$

Now, let $G(t, s_1, s_2)$ be the probability generating function (p.g.f.) of the probability density $P_{x_1, x_2}(t)$, i.e.

$$(3.5) \quad G(t, s_1, s_2) = \sum_{x_1=0}^{\infty} \sum_{x_2=0}^{\infty} P_{x_1, x_2}(t) s_1^{x_1} s_2^{x_2}.$$

By straightforward algebraic manipulations it can be shown that as $\Delta t \rightarrow 0$, equations (3.1) – (3.4) give the following differential equation for $G(t, s_1, s_2)$:

$$(3.6) \quad \frac{dG}{dt} = -(\lambda_1 + \lambda_2 - \lambda_{12})G + (\lambda_1 - \lambda_{12})s_1G + (\lambda_2 - \lambda_{12})s_2G + \lambda_{12}s_1s_2G.$$

Solving (3.6) with the initial condition $G(0, s_1, s_2) = 1$, we get

$$(3.7) \quad G(t, s_1, s_2) = \exp [-(m_1 + m_2 - m_{12}) + (m_1 - m_{12})s_1 + (m_2 - m_{12})s_2 + m_{12}s_1s_2],$$

where $m_1 = \lambda_1 t$, $m_2 = \lambda_2 t$, $m_{12} = \lambda_{12} t$. Picking out the coefficient of $(s_1^{x_1} s_2^{x_2})$ in (3.7), we find

$$(3.8) \quad P_{x_1, x_2}(t) = \exp [-(m_1 + m_2 - m_{12})] \sum_{i=0}^{\min(x_1, x_2)} \frac{(m_1 - m_{12})^{x_1 - i}}{(x_1 - i)!} \cdot \frac{(m_2 - m_{12})^{x_2 - i}}{(x_2 - i)!} \cdot \frac{m_{12}^i}{i!}$$

where the upper limit of summation is $\min(x_1, x_2)$. (3.8) is the usual probability density function of the bivariate Poisson distribution with marginal parameters m_1 , m_2 and correlation coefficient $m_{12}/(m_1 m_2)^{1/2}$.

Finally, a direct generalization of the method of this note produces a multivariate Poisson process, which leads to an axiomatic derivation of the multivariate Poisson distribution [Krishnamoorthy (1951), Teicher (1954)].

REFERENCES

- CAMPBELL, J.T. (1934). The Poisson correlation function. *Proc. Edinburgh Math. Soc.* (Ser. 2), 4, 18-26.
- DOOB, J.L. (1953). *Stochastic Processes*. John Wiley & Sons, Inc., New York.
- HAMDAN, M.A. and AL-BAYYATI, H.A. (1969). A note on the bivariate Poisson distribution. *The American Statistician*, 23, 32-33.
- HAMDAN, M.A. and TSOKOS, C.P. (1971). A model for physical and biological problems: The bivariate compounded Poisson distribution. *Review of the Int. Statist. Inst.*, 39, 60-63.
- KRISHNAMOORTHY, A.S. (1951). Multivariate binomial and Poisson distributions. *Sankhya*, 11, 117-124.
- MARITZ, J.S. (1950). On the validity of inferences drawn from the fitting of Poisson and negative binomial distributions to observed accident data. *Psychological Bulletin*, 47, 434-443.
- TEICHER, H. (1954). On the multivariate Poisson distribution. *Skand. Aktuarietids.*, 37, 1-9.

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